Caution Lights Dimming Control Box – General Data

1. General

The Caution Lights Dimming Control Box forms part of the Master Warning and Master Caution system, and consists of a remote mounted light dimmer control box which is capable of controlling up to 12 independent lamps with full test functions.

2. Description

A three-position switch labelled CAUTION LIGHTS with positions DIM, BRIGHT, and TEST is located on the lower left sub-panel. The momentary TEST position of the switch permits a simultaneous check of Autofeather, Master Caution, Master Warning, Stall and Fire lights. The Fire Lights are tested using their own dedicated switch. The test and intensity circuit is protected by a CAUTION LTS NORMAL avionics breaker on the main circuit breaker panel. The DIM and BRIGHT positions provide alternative levels of lighting brilliance for the MASTER CAUTION and MASTER WARNING lights, the autofeather system SELECT and ARMED indicator lights, and the stall warning light, and the engine FIRE warning lights. In the event of failure of the control system used for caution light dimming, a reversionary, non-dimming mode can be selected by moving the CAUTION LIGHTS DIMMING switch on the lower right sub-panel from NORMAL to BYPASS. It is powered by the CAUTION LTS BYPASS circuit breaker on the avionics circuit breaker panel.

3. Adjustment/Test

A. Apply DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

B. Caution Lights (Bright)

1) Press the CAUTION LIGHTS to TEST and verify that the following lights illuminate:

a) AUTOFEATHER (L & R)

b) MASTER CAUTION (L & R)

c) MASTER WARNING (L & R)

d) STALL

e) FIRE (L & R)

C. Caution Lights (Bypass)

1) Pull the CAUTION LTS NORMAL circuit breaker (3N).

2) Select BYPASS on the CAUTION LIGHTS DIMMING switch.

3) Press the CAUTION LIGHTS to TEST and verify that the following lights illuminate:

a) MASTER CAUTION (L & R)

b) MASTER WARNING (L & R)

4) Operate the AUTOFEATHER switch to illuminate SELECT and verify illumination.

5) Operate the FIRE WARNING switch to illuminate FIRE (L & R) and verify illumination.

6) Operate the sensor on the wing to illuminate the STALL light.

7) Reset the CAUTION LTS NORMAL circuit breaker (3N).

D. Caution Lights (Dim)

1) Autofeather (L & R):

a) Set the CAUTION LIGHTS to BRIGHT.

b) Operate the AUTOFEATHER switch to illuminate SELECT.

c) Set the CAUTION LIGHTS to DIM and verify a reduction in brightness.

2) Master Caution (L & R) & Master Warning (L & R):

a) Set the CAUTION LIGHTS to BRIGHT.

b) Pull the DAU 2B circuit breaker to trigger the MASTER CAUTION (L & R) and MASTER WARNING (L & R).

c) Set the CAUTION LIGHTS to DIM and verify a reduction in brightness.

d) Reset the DAU 2B circuit breaker.

3) Fire (L & R):

a) Set the CAUTION LIGHTS to BRIGHT.

b) Operate the FIRE WARNING switch to illuminate FIRE (L & R).

c) Set the CAUTION LIGHTS to DIM and verify a reduction in brightness.

4) Stall:

a) Set the CAUTION LIGHTS to BRIGHT.

b) Operate the sensor on the wing to illuminate the STALL light.

c) Set the CAUTION LIGHTS to DIM and verify a reduction in brightness.